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MIL-STD-331B
NOTICE 5
1 August 1995

MILITARY STANDARD
FUZE AND FUZE COMPONENTS
ENVIRONMENTAL AND PERFORMANCE TEST FOR

TO ALL HOLDERS OF MIL-STD-331B:

1. THE FOLLOWING PAGES OF MIL-STD-331B HAVE BEEN REVISED AND SUPERSEDED THE PAGES LISTED:

NEW PAGE	DATE	SUPERSEDED PAGE	DATE
v	1 August 1995	v	1 May 1994 (Notice 4)
vi	1 May 1994	vi	REPRINTED WITHOUT CHANGE
vii	1 August 1995	vii	1 December 1992 (Notice 3)
viii	1 December 1989	viii	1 May 1994
F2-1 thru F2-11	1 August 1995	NONE	NEW PAGES
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2. RETAIN THIS NOTICE AND INSERT BEFORE TABLE OF CONTENTS.

3. Holders of MIL-STD-331B will verify that page changes and additions indicated above have been entered. This notice page will retained as a check sheet. This issuance, together with appended pages, is a separate publication. Each notice is to be retained by stocking points until the military standard is completely revised or canceled.

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TEST F2

HIGH-ALTITUDE ELECTROMAGNETIC PULSE

F2.1 PURPOSE. This is a laboratory test which determines fuze ability to satisfy safety and reliability requirements when exposed to a simulated high-altitude electromagnetic pulse (HEMP) environment. This HEMP potentially could initiate or alter electro-explosive devices (EEDs) and destroy or damage vulnerable electronic components in the fuze.

F2.2 DESCRIPTION

F2.2.1 General. This test is intended to evaluate fuzes in their storage, transportation and handling configurations. Testing of fuzes with munitions in their tactical configurations is discussed under Section F2.6. Fuzes shall be subjected to one of the simulated electromagnetic pulse environments described below. Where the test capability is available, the classified HEMP environment shall be utilized. These environments are applicable to altitudes less than 20,000 meters.

F2.2.1.1 Unclassified HEMP. The unclassified HEMP (see F2.7.6.1) is defined as:

$$E(t) = 5.25 \times 10^4 (e^{-4\times10^6 t} - e^{4.76\times10^8 t})$$

and

$$H(t) = E(t) \div z$$

where

t is time in seconds;

z = 377 ohms, the characteristic impedance of free space;

E is the electric field in volts/meter; and

H is the magnetic field in amperes/meter.

The above equation has a pulse wave shape with time, with a peak field strength (Ep) of 50 kilovolts per meter, a 10 to 90 percent rise time of 4.15 nanoseconds, a time to half-value of 185.5 nanoseconds and a time duration between .1 Ep values of 587.8 nanoseconds. The Electric Field Spectrum and Normalized Cumulative Energy Density Spectrum are presented in Figure F2-1 (see F2.7.6.2). The minimum requirements for the simulation of this

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waveform for the purpose of this test are presented in F2.4.1.3. It should be noted that this unclassified waveform may not be suitable for final evaluation.

F2.2.1.2 Classified HEMP. The secret HEMP environment is defined in MIL-STD-2169.

F2.2.2 Fuze configuration. The fuze shall be completely assembled, including all electronic circuits and electroexplosive devices that are a part of the fuze design. Lead and booster charges may be omitted to facilitate testing, if insensitive or inaccessable to HEMP, or substituted with appropriate inert explosive component simulant provided that the electromagnetic configuration of the munition is maintained. If the fuze is normally shipped or stored on a munition, then the test shall be conducted with the fuze assembled in an inert test munition that provides the same electromagnetic characteristics, such as conductivity, shielding, internal and external wiring of the operational munition. If the fuze is used in multiple munitions, the test shall be conducted in each munition or the munition in which the fuze is considered to be the most susceptible to HEMP. If the fuze is not normally installed in a munition for shipping or storage, then the fuze in its standard packaged configuration shall be tested. If the fuze is normally shipped or stored unpackaged, it shall be tested in the unpackaged configuration. The fuze configurations chosen for the test shall be based on all known transportation, storage and handling configurations and the susceptibility analyses as described in F2.2.5.1 of this test; see also F2.7.4.3.

F2.2.3 Applicable publications. All standards, specifications, procedures and manuals which form a part of this test are listed in Section 2 of this standard.

F2.2.4 Number of test items

a. If the item is to be instrumented, a single test item is sufficient for each individual test sequence (i.e., operational mode, test configuration and orientation). Multiple items should be tested to assess round-to-round variations in hardness.

b. If the item is not to be instrumented, a minimum of 10 items is required for each individual test. Each individual test will include exposure to at least 10 HEMP pulses. For tests where 10 items are not available, see F2.3.2.2.

F2.2.5 Documentation. Test plans, performance records, equipment, conditions, results, and analyses shall be documented in accordance with section 4.8 of the general requirements of this standard. The following unique requirements also apply.

F2.2.5.1 Analyses. The HEMP coupling analyses shall be conducted for all known transportation, storage and handling configurations for the fuze. The analyses should determine and provide the most significant system configurations, test configurations and orientations; whether the fuze is to be instrumented for test; the determination of the parameters to be monitored; the expected stress levels; the component thresholds for upset and permanent damage; prioritization of likely failure modes and rationale for the components to be instrumented; etc

F2.2.5.2 Test plan. The formulation of an appropriate test plan shall be based on the analyses of F2.2.5.1. The test plan shall include:

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a. Identification of the fuze items to be tested at the applicable level of component integration (i. e., system, munition, fuze, subsystem, assembly, circuit, individual component, etc.), and the following pertinent data and information:

1) The tests and parameters to be measured before, during and after the HEMP environment is applied. The test record data sheet format shall be included.

2) The test points and supporting rationale for choices.

3) Instrumentation employed for the response measurements.

4) Ambient conditions.

5) Functional modes to be evaluated and supporting rationale for choices.

6) Number of test items and controls:

a) See F2.2.4 and F2.3.2.2.

b) The time between pulses shall be greater than 5 times the thermal time constant of the components of concern.

7) Physical configuration of the test items and any ancillery equipment when exposed to the test environment, and the number of test items for each configuration, operational mode and orientation.

8) The number and kinds of spare parts required.

9) The data to be recorded.

10) The specified HEMP free field test waveforms and frequency spectra.

11) The method of operation and monitoring of the equipment.

b. A description of the test facilities to be employed to include instrumentation and simulator characteristics, environment measurement techniques, and calibration procedures. Also the justification for choosing the simulator or simulation techniques, including the pertinent simulator characteristics needed.

c. A description of how the chosen simulator environments resemble and differ from the threat environment and the methodology for extrapolating the test results to those that would result by exposing the system to the threat environment.

d. A statement of the specific test levels and test sequences and the survivability requirements to be assigned to each test.

e. The controls to protect personnel and equipment in event the fuze functions during test.

F2.2.5.3 Test report. The test report shall contain the test plan, all the data and the conclusions resulting from the tests delineated in the test plan. In particular the test report shall provide:

a. The transient responses of the sensor outputs, starting with time zero, as well as that of the simulated HEMP field.

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b. A statement of how the simulated environments were measured; where with respect to the sample the measurements were made; and what device or instruments were used.

c. A detailed description of the instrumentation calibration procedures.

d. A description of the operational steps used to set up the fuze for tests.

e. A description of how the actual test procedure differed from the test plan.

f. Details of the fuze post-exposure status.

g. A detailed description of how the response information on components, subassemblies and assemblies is analyzed to arrive at an evaluation of the system survivability. This analysis should relate measured data to the components' failure levels.

h. A statement of whether the fuze has met the criteria for passing the test (Section F2.3) and the rationale for this conclusion including assumptions and engineering judgments made to support the conclusions.

F2.3 CRITERIA FOR PASSING TEST.

F2.3.1 Fuze condition. At the completion of this test, the fuze shall be safe for transportation, storage, handling and use, as well as operable in accordance with Paragraphs 4.6.2.1 and 4.6.2.2a of the general requirements to this standard. The fuze shall perform in accordance with its performance specifications.

F2.3.2 Decision basis. Breakdown, inspection, other appropriate tests and engineering judgment shall form the basis for the decision that fuzes have passed or failed the test.

F2.3.2.1 Instrumented fuzes. For electroexplosive devices, the maximum pin-to-pin and pin-to-case no-fire criteria establish the baseline. For electronic components, threshold damage data based on actual tests or analyses may be used; in their absence, Wunsch damage criteria apply. See references F2.7.6.3 and F2.7.6.4. For all components for which a failure represents a safety hazard, a 20 dB safety factor shall be applied to induced power, energy, voltage and/or current,depending on the damage characteristics of the component. For all other components, a 10 dB safety factor shall be applied.

F2.3.2.2 Non-instrumented fuzes. For tests that have used at least 10 items exposed 10 times for each individual test, a test stimulus amplitude of at least threat level is required plus a 10 percent margin to account for environmental measurement errors. For tests that have used less than 10 items, but at least two, a test stimulus amplitude of at least 6dB above threat level is required.

F2.4. EQUIPMENT.

F2.4.1 HEMP simulator. A HEMP simulator such as operated by the Army, Navy or Air Force may be used. The services have used bounded wave simulators as shown in Figure F2-2, large enough to accommodate a large airplane and others small enough to sit on a

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laboratory work bench. Other simulators meeting the specifications below may also be used. (see F2.7.3).

F2.4.1.1 Field intensity uniformity. With the test munition outside the test area, the field intensity in the test area shall be measured at 6 equally spaced points on the surface that would be generated by rotating the test munition about its center in all directions; the center of rotation will be at the center of the test area. The position of test points shall insure that only one line connected between opposite points will be parallel to the E-field. The field intensities measured on the surface shall not differ by more than 10% from those measured at the center of rotation.

F2.4.1.2 Field intensity loading. It must be ensured that the test munition does not excessively load the test fields. This would significantly alter the induced currents on and in the munition and, therefore, invalidate the test. For a bounded wave simulator, this requires that the ratio of the largest dimension of the test munition to the plates' separation distance be 0.6 or less.

F2.4.1.3 Unclassified HEMP. The equipment shall provide HEMP fields as specified in Figure F2-3 with a risetime of 10 ns or less, a fall time greater than 100 ns and a peak aplitude of at least 50 kV/m.

F2.4.1.4 Secret HEMP. The equipment shall provide HEMP fields as specified in MIL-STD-2169.

F2.4.2 Fuze instrumentation. The fuze may be instrumented to measure induced currents and/or voltages at the critical circuit locations determined by the analyses of F2.2.5.1. For EEDs, both pin-to-pin currents and pin-to-case voltages shall be monitored. Fuze instrumentation shall not excessively distort the incident field, induce spurious signals in the fuze or alter the data. This may be accomplished with fiber optics or microwaves to transmit data between the fuze under test and the remotely positioned recording instrumentation. Alternately, data can be brought out via well-shielded coaxial lines with connectors in the fuze skin. The lines should be routed perpendicular to the E-field using appropriate RF-absorbant material to minimize electromagnetic coupling. Tests should be performed to insure there are no instrumentation-induced errors.

F2.5 PROCEDURE

The fuze in its most vulnerable configurations, as described in F2.2.2, shall be exposed to the simulated HEMP. For all test configurations, the fuze shall be so oriented with reference to the incident E-field and H-field as to ensure maximum coupling of energy to all components of concern. If maximum coupling to different components requires different orientations, or if worst case coupling cannot be determined a priori, various orientations shall be used to assure testing in the worst case conditions.

F2.5.1 Test arrangements. The arrangement that gives maximum coupling shall be used if known; otherwise, both of the following arrangements shall be used.

F2.5.1.1 Monopole. The fuze under test shall be supported by a dielectric structure. One end of the munition shall be grounded, thereby simulating a monopole antenna. The major axis of this arrangement shall be oriented parallel to the electric field.

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F2.5.1.2 Dipole. The fuze under test shall be supported by a dielectric structure and isolated from ground, thereby simulating a dipole antenna. The major axis of this arrangement shall be oriented parallel to the electric field.

F2.5.2 Compliance. The fuze shall be tested according to the test plan and the results analyzed to determine whether the fuze meets the pass/fall criteria in Section F2.3.

F2.6 ALTERNATE AND OPTIONAL TESTS.

For those instances where the fuze developer must evaluate the fuze vulnerability to HEMP under tactical conditions, tests shall be performed with the fuzes and munitions in their launch preparation, launch, and flight configurations as applicable. All fuze modifications, instrumentation, analyses, and documentation requirements are identical to those for tests performed in the transportation, storage and handling configurations.

F2.7 RELATED INFORMATION.

F2.7.1 HEMP generation. A nuclear detonation creates an intense Electromagnetic Pulse (EMP) in addition to the shock, blast, ionizing radiation, and thermal effects. Although EMP is generated by surface, air, and high altitude nuclear bursts, it is the primary weapons effect of a high-altitude (exoatmospheric) burst and illuminates the largest area on the earth' s surface, thousands of square miles. In addition, at distances great enough for the other effects to become small, the high-altitude burst generates the most severe EMP (HEMP) threat. For a high-altitude nuclear burst occurring at altitudes above 40 km, gamma rays collide with air molecules (Compton collisions), causing electrons to be ejected. These Compton recoil electrons spiral under the influence of the earth's magnetic (geomagnetic) field. Spiraling causes them to accelerate, and hence radiate, thereby producing the electromagnetic pulse.

F2.7.2 HEMP Characteristics.

F2.7.2.1 Origin of the unclassified HEMP. High-altitude EMP develops significant field strength over a wide area of coverage. However, its amplitude-time history and polarization are not uniform over this area, and depend primarily on the orientation of the geomagnetic field and the observer's location in comparison to the burst. A double exponential generalized wave shape, independent of observer location, has been developed by the EMP community, and is used as a working characterization (see F2.2.1.1 and Figure F2-3). This constructed wave form represents a composite of extremes that occur in different regions within the area of coverage, combining the shortest rise time, maximum peak amplitude and longest fall time. Because of the geomagnetic dip angle, over the United States, the HEMP is primarily horizontally polarized with a peak electric field amplitude of 50,000 v/m, with a vertical component of 15,000 v/m. This represents the incident fields. The total fields at any point also include reflection of these fields from the ground plane. The total fields can be larger or smaller than the incident fields, depending on polarization, These characteristics are the basis of worst case evaluation of HEMP interaction with fuzes.

F2.7.2.2 Comparison with other EM fields. HEMP is similar to the radiated electromagnetic pulse produced by a lightning stroke. However, it has a faster rise time and illuminates a much larger area with an intense field. The HEMP electric field intensity rises to levels as high 50kv/m in less than ten nanoseconds, and then decays to an insignificant value

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in less than 1.0 microsecond. Although this time span is very short, the pulse amplitude is much larger than ordinary radio signals. When HEMP is coupled into an object, the object "rings" at one or more resonant frequencies. The current or voltage wave forms induced are thus usually a superposition of damped sinusoids. The ability of HEMP to couple energy to an object often depends strongly on its spectral amplitude at these resonances. In addition to direct excitation of system elements, HEMP energy can be transferred to a munition system by the interconnecting cables which act as receiving antennas and transmission lines.

F2.7.3 Simulation techniques.

F2.7.3.1 Use of HEMP simulators. It is not feasible to explode a nuclear bomb whenever the need to assess the susceptibility of a weapon to HEMP arises. Hence HEMP simulators have been developed to simulate the HEMP environment. As with most environmental simulators, each HEMP simulator has particular strengths and weaknesses with regard to simulating the actual HEMP, depending on the test conditions and the weapon to be tested. Extensive knowledge about the system to be tested and effects to be examined are necessary for selecting a HEMP simulator.

F2.7.3.2 Types of HEMP simulators. Many kinds of HEMP simulators have been developed and are in the process of being developed. Simulation test techniques are similarly in their evolutionary phases. Recent concerns of the Environmental Protection Agency (EPA) about HEMP simulators polluting the environment with their fields has stimulated interest in those which have minimum radiated fields. Also, simulators that cost less to purchase, operate and maintain have become more widely used, adding to their consideration for use as a standard. The equipment and techniques given below are discussed in greater detail in references F2.7.6.5 and F2.7.6.6.

F2.7.3.2.1 Pulse radiation simulators. Biconic Dipole and Resistive Loaded Horizontal Dipole (both horizontally polarized), and Inverted Conical Monopole (vertically polarized) are all classified as pulse radiation simulators. The intensity of the useful test fields that can be radiated by these devices is limited because of the following factors:

a. Test field intensity falls off inversely with distance.

b. Test object must be positioned a sufficient distance from the antenna to assure far-field conditions.

c. Maximum field intensity that can be generated is limited by the state-of-the-art in the design and manufacture of high voltage generators and switches and by their high cost to purchase and maintain.

d. Field intensity uniformity cannot be maintained across large systems placed close to the simulator.

F2.7.3.2.2 Bounded wave transmission line simulators. These simulators can be used to establish plane waves within the two plates of a transmission line as depicted in Figure F2-2; this provides vertically polarized fields. Another type makes use of two vertically positioned plates positioned above ground, thereby, providing balanced lines. This type is horizontally polarized and has the advantage of minimum radiated fields, with the external fields at a distance 100 feet from the chamber being down more than 60 dB from those inside

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the chamber. A third type, called the Crawford cell, was developed by W. Crawford, National Institute of Standards and Technology. This has a three-dimensional shape similar to the parallel plate lines, except it is a coaxial system enclosed on all sides. This cell has the advantage of radiating even less than the balanced line, but for a given size has less test area available. Because the energy delivered to these transmission lines is more confined than from radiating antennas, the same intensity energy source can establish much higher intensity fields. Where the plates are closer together, the fields can be proportionately higher. It is possible to design a chamber to provide the choice of a 377 ohm field, a low impedance field or a high impedance field. This allows simulation of a reflected field that adds or subtracts from' the E-field component. These chambers can be used to establish either HEMP fields or CW fields. The maximum HEMP field available from an existing chamber is 125 kV/m; in this case the separation distance between plates is 40 feet. Moving the plates closer increases field intensity; increased field intensity for smaller objects can be obtained in the transition section of this chamber.

F2.7.3.3 Low intensity test fields. Low intensity test fields can be useful in the developmental stage of an item to evaluate weaknesses in the system and to aid in design. However, it is not possible to extrapolate low intensity field effects up to threat levels because of weapon component non-linearities at those levels. Final evaluation of the system is necessary at threat level fields or their equivalents.

F2.7.4 Reasons for HEMP concern. The increasing dependence of military operations on sensitive and sophisticated electronic equipment and the large-scale introduction of semi-conductor devices have significantly increased the possibility of collateral damage and degradation of mission performance due to HEMP. An area of particular concern is the HEMP vulnerability of ordnance. Potential exists for sufficient HEMP energy to be coupled to munitions to initiate EEDs. Fuzes having electronic safety and arming devices with in-line explosive trains are a more serious safety concern than those having conventional, interrupted explosive trains; for the latter, reliability is the primary concern. Furthermore, with the advent of smart weapons, much of the electronics, especially digital logic circuits responsible for propulsion control, arming, and fusing and guidance subsystems, are potentially susceptible to HEMP-induced damage or upset.

F2.7.4.1 EED initiation. In the case of ordnance containing sensitive EEDs, the HEMP energy could cause initiation. The initiation of EEDs could cause a safety hazard for ships, aircraft, weapon systems and personnel by directly functioning the fuze or its safety features or cause the dudding of ordnance. The mechanisms for initiation of the EED could include electrical breakdown through the explosive material between the pin and case as well as pin-to-pin current. Even when the heat generated by the single pulse of energy may be insufficient to initiate an EED, the EED may become permanently desensitized as a result of this thermal influence and fail to function properly at a later time. This latter case presents a reliability problem as the ordnance is effectively dudded.

F2.7.4.2 Electronic system damage or upset. Induced HEMP transients can produce two types of detrimental responses in electrical/electronic equipment: upset and permanent damage. Upset is the temporary generation of false signals that cause the system to take undesired actions. Damage refers to the degradation of a component to the point where it can

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not perform its design function. Both safety and reliability problems can result from either type of response.

F2.7.4.3 Other areas of concern. As subsystems are assembled into a complete operating system, their susceptibility to HEMP can change dramatically due to changes in effective shielding and size of the inadvertent antenna system exposed to HEMP. HEMP tests should, therefore, include evaluation of the vulnerability of the fuze in all of its significant system configurations. Where called for, tests should include munition operational environments before, during, and after launch.

F2.7.5 Test applicability.

a. The HEMP test is primarily a fuze test and need only be applied to those fuzes that contain electroexplosive devices or other electrical/electronic components. All fuzes should be safe and reliable following exposure to HEMP while they are in their storage and transportation configurations to ensure against a complete loss of capability of the stockpile in case of an HEMP event. If the fuze is also stored or transported while mounted on one or more munitions or outside its storage container, then these are appropriate test configurations. These tests will not evaluate electronic system upset, since this normally requires the electronic circuits to be activated, a condition that will not exist in transportation and storage configurations. All these tests are the responsibility of the fuze developer.

b. A statement, similar to a. above, can be made for the munition without fuze attached. Although these tests are normally the responsibility of the munition developers, test F2 may be used in the absence of a specific HEMP test for unfuzed munitions.

c. In general, for a munition, the conguration most susceptible to HEMP occurs during preparations for launch. At this time cables entering the munition, create antennas capable of capturing energy from electromagnetic fields. Also, the electronics and guidance system are activated. The second most susceptible configuration is the munition in flight, again because the electronics are activated, the fuze may be armed, and there may be an electrically conductive exhaust plume adding to the inadvertent antenna systems and increasing the chances for a fuze premature or malfunction. Not all munitions are required to be safe from HEMP under these conditions, because of practicality limitations.

d. For those munitions or weapon systems for which HEMP survivability is essential to mission success or platform safety, HEMP vulnerability tests are required. These tests should be conducted in all significant munition/weapon systems handling configurations to evaluate effects on EEDs, electronic system damage and upset on all components in the system, including those in the fuze. Test F2 does not necessarily apply to complete system tests and responsibility normally lies with the munition weapon developer.

F2.7.6 Bibliography.

F2.7.6.1 Defense Nuclear Agency EMP Handbook.

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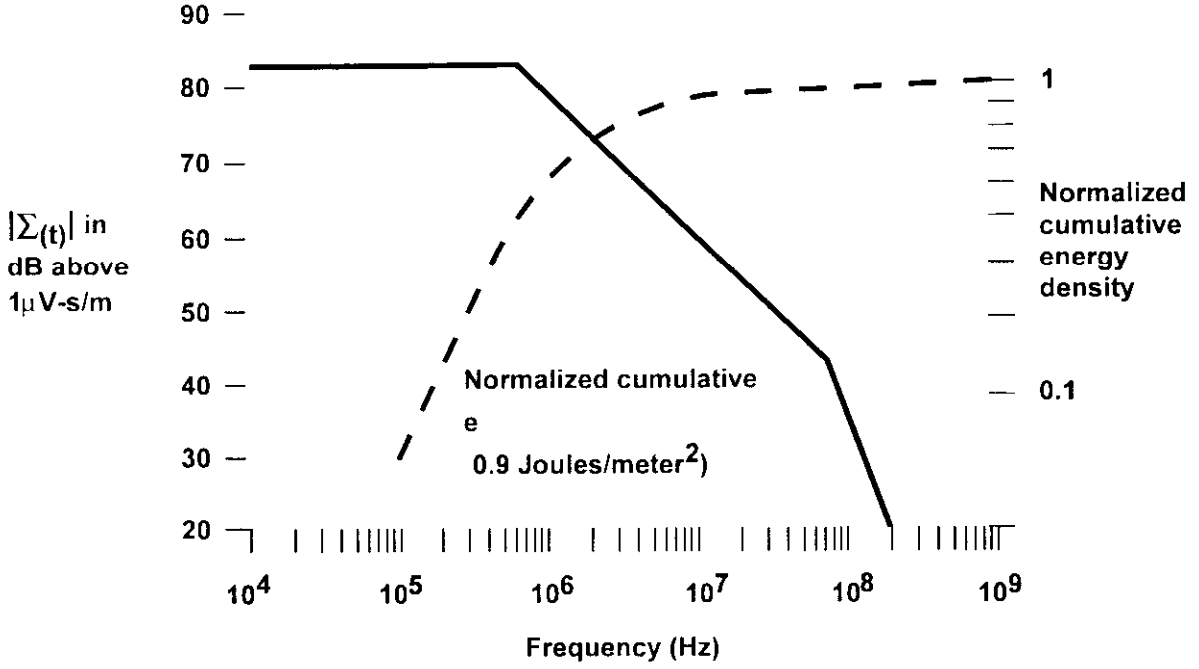


FIGURE F2-1. High-Altitude EMP Spectrum and Normalized Energy Density Spectrum

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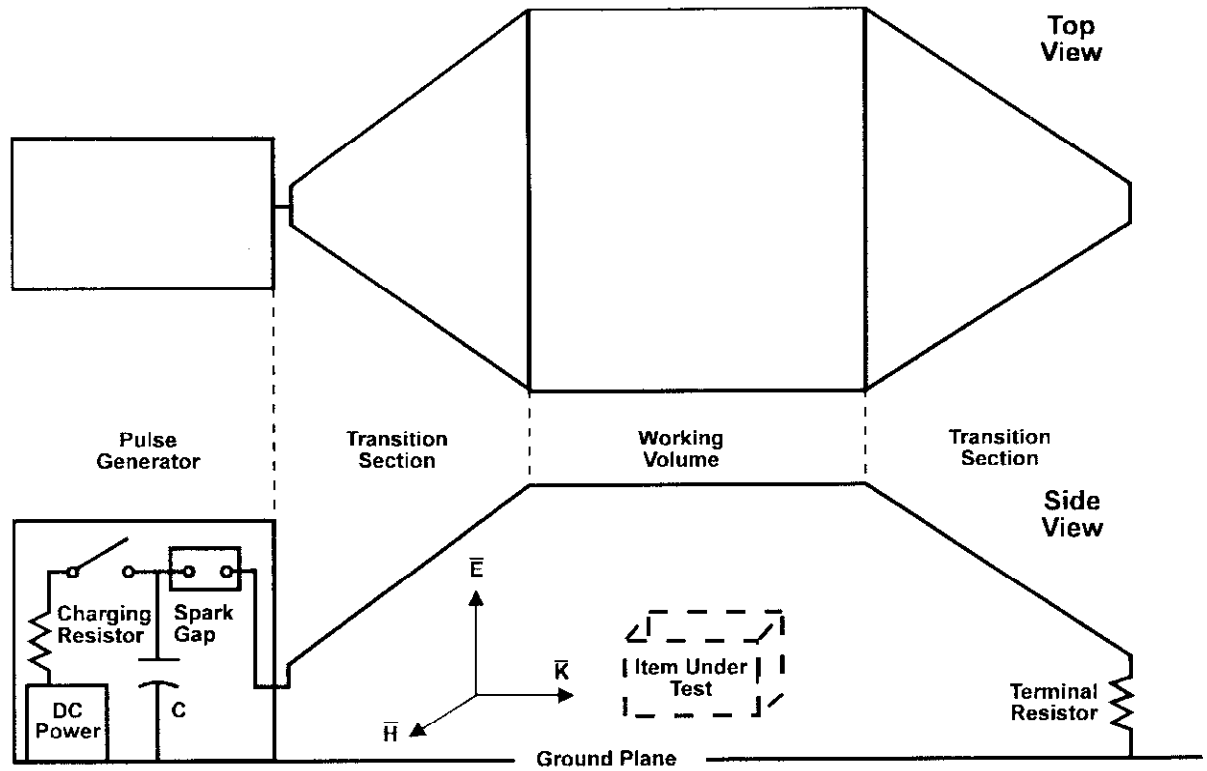


FIGURE F2-2. Bounded Wave Simulator

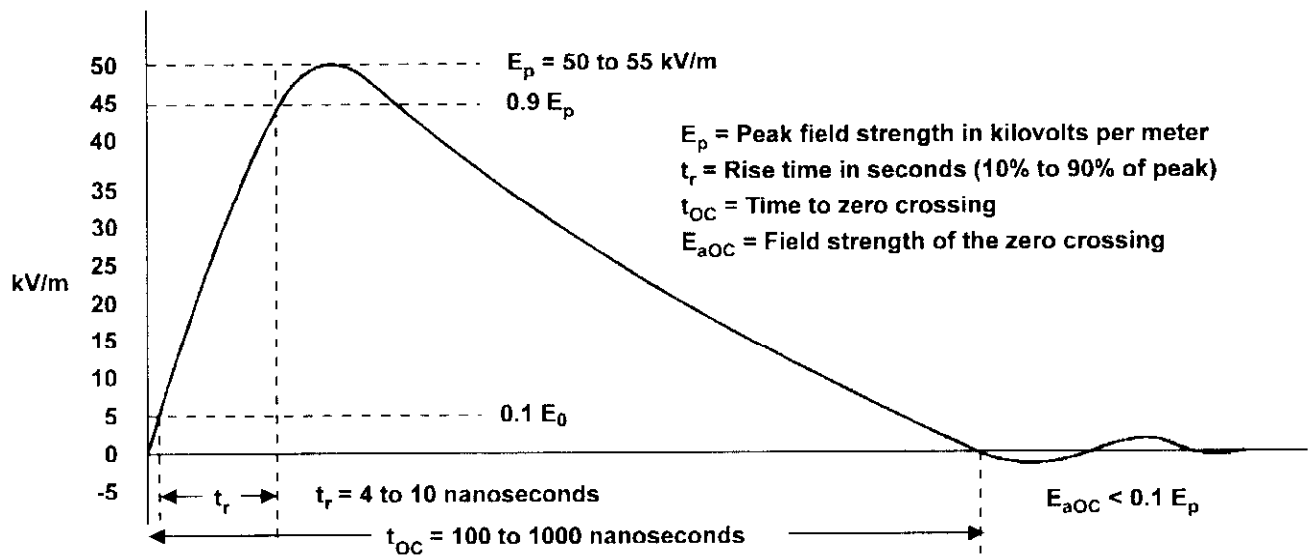


FIGURE F2-3. Characteristics of UNCLASSIFIED HEMP Simulator

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TEST F3

ELECTROMAGNETIC RADIATION HAZARDS (EMRH)

F3.1 PURPOSE. This is a laboratory safety and reliability test simulating Electromagnetic Radiation (EMR) which may impinge upon the fuzes containing Electro-explosive Devices (EEDs) during their life cycle. Fuze EEDs must withstand the high levels of electromagnetic radiation which may be encountered during storage, transportation, handling, loading and launching.

F3.2 DESCRIPTION.

F3.2.1 General. This test evaluates the effect of subjecting bare and packaged unarmed fuzes to high EMR environments. Bare fuzes are exposed to the EMR environment which they are expected to encounter during handling, loading and launching. Packaged fuzes are exposed to the EMR environment which they are expected to encounter during storage and transportation.

F3.2.1.1 Storage and transportation EMRH test. This test shall be conducted on packaged fuzes to evaluate their safety and reliability during and after exposure to a high EMR environment while in storage and transportation.

F3.2.1.2 Handling, loading and launching EMRH test. This test shall be conducted on bare fuzes to evaluate their safety and reliability when exposed to a high EMR field during handling, loading and launching.

F3.2.2 Electromagnetic environments.

F3.2.2.1 Storage and transportation EMR environment. Service requirements for storage and transportation EMR environments vary somewhat though MIL-STD-1385(Navy) is generally accepted. Army procured items have an additional requirement to withstand a minimum of 200 Vrms/m at all frequencies.

F3.2.2.2 Handling, loading and launching EMR environment. Service requirements vary widely for EMR during handling, loading and launching of munitions. The Navy and Marines require MIL-STD-1385 environments for launching, and use personnel exposure limits for handling and loading. The Army Aviation community has an additional requirement of 200 Vrms/m minimum at all frequencies. Most Army procured munitions have a lower handling, loading and launching environment which is listed in Table F3-I.

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F3.2.3 Fuze configuration. The fuzes shall be completely assembled except that lead and booster charges may be omitted to facilitate testing. If any explosive elements are removed, care should be exercised to preserve electromagnetic equivalency of the resulting configuration. All EEDs shall be replaced with appropriate instrumentation to measure the effect of the EMR environment on the EED.

TABLE F3-I. Army Electromagnetic Radiation Environment During Handling, Loading and Launching Munitions

Frequency (MHz)	Electric Field (Vrms/m)		Polarization
	Average	Peak	
0.1 to 2	50	100	Vertical
2 to 32	100	200	Vertical
32 to 100	100	200	Vertical and horizontal
100 to 1,000	200	400	Vertical and horizontal
1,000 to 18,000	200	20,000 *	Vertical and horizontal

* Design goal; not a test requirement.

F3.2.4 Applicable publications. All standards, specifications, drawings, procedures and manuals which form a part of this test are listed in Section 2 of the introduction to this standard. Special attention is directed to MIL-STD-1512, MIL-STD-1385, MIL-I-23659, TOP 1-2-511, and ADS-37 which have specific applications.

F3.2.5 Number of test items. A single test item is sufficient for an instrumented EMRH test. However, more than one may be required to facilitate instrumentation.

F3.2.6 Test documentation. Test plans, performance records, equipment, conditions, results, and analysis shall be documented in accordance with Section 4.8 of the general requirements to this standard. The following unique requirements also apply.

F3.2.6.1 Analyses. EMR coupling analyses shall be performed for all known storage, transportation, handling, loading and launch configurations for the fuze. The analyses should determine and provide the most significant life cycle configurations, test configurations and orientations; the type of fuze instrumentation to be used for the test; the determination of the parameters to be monitored; the expected stress levels and the 0.1% probability 95% confidence no-fire characteristics of all EEDs.

F3.2.6.2 Test plan. The formulation of an appropriate test plan shall be based on the analysis of F3.2.6.1. The test plan shall include:

a. identification of the fuze items to be tested at the applicable level of component integration (i.e., system, munition, fuze, subsystem, etc.), and the following pertinent information:

   1) The physical condition of the fuze items to be tested.

   2) The test points and supporting rationale for choices.

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3) A description of the instrumentation installed in the fuze for response measurements and the minimum sensitivity requirements to ensure the appropriate safety factors can be demonstrated in the test EMR environment.

4) The specific data to be recorded.

5) The method of operation and monitoring of the equipment.

b. A description of the test facilities to be employed to include instrumentation and transmitter characteristics, environment measurement techniques, and calibration procedures.

c. A description of the test environment including field intensity, polarization, frequency range, number of test frequencies, rationale for frequency selection, and any modulation characteristics employed.

d. A description of how the test environment differs from the threat environment and the methodology for extrapolating the test results to those that would result by exposing the system to the threat environment.

e. A description of the specific procedures to be utilized during the test including the configuration of the test items, their orientation(s) with respect to the test field, the length of time of each exposure to an EMR environment and the data recording procedure.

F3.2.6.3 Test report. The test report shall contain the analyses of F3.2.6.1; the test plan; and all the raw data, reduced data, results and conclusions resulting from the tests delineated in the test plan. In particular the test report shall provide:

a. The responses of the EED instrumentation to the EMR environment.

b. A statement of how the test environments were measured including the type of field probes used and placement of the probes with respect to the fuze tested.

c. A detailed description of the instrumentation calibration procedures and complete calibration data for all sensors used to monitor EED responses.

d. A description of how the actual test procedures differ from those in the test plan.

e. A detailed description of how the raw data was analyzed and compared with EMR environment characteristics and EED no-fire characteristics to determine what safety factors were achieved.

f. A summary of results to include a presentation of the worst case safety factors for each configuration tested. Graphical representation of the data is encouraged to increase the clarity of results.

g. A statement of what conclusions can be drawn from the results regarding the safety and reliability of the fuze when exposed to the EMR environments to be encountered during its life cycle.

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F3.3 CRITERIA FOR PASSING TEST.

F3.3.1 Fuze condition. The fuze EEDs must remain safe and reliable when exposed to the appropriate EMR environments to be encountered during storage, transportation, handling, loading, and launching.

F3.3.2 Decision basis. An analysis of the test data will form the basis for determining if the fuze has passed or failed this test. Data shall be analyzed for each configuration to determine if the following criteria is met:

a. For EEDs within fuzes configured in a manner that their premature function will cause the initiation of a main charge or removal of a safety feature, no more than the no-fire characteristics of that EED reduced by 16.5 dB shall be induced in it due to exposure to the specified EMR environment.

b. For EEDs within fuzes configured in a manner that their premature function will cause the munition to fail to function reliably, no more than the no-fire characteristics of that EED reduced by 6.9 dB shall be induced in it due to exposure to the specified EMR environment.

F3.4 EQUIPMENT.

F3.4.1 Transmitter. The transmitting equipment used for the test must have sufficient stable power output over the EMR environment frequency range to ensure that appropriate safety factors can be verified. Frequency output should be controllable to within a nominal 2% of each desired test frequency. Laboratory transmitting equipment normally consists of a series of RF signal generators and wideband power amplifiers which amplify the output of the signal source to hundreds or thousands of watts. Some U.S. military test facilities have transmitting equipment with a peak power capability exceeding 100,000 watts.

F3.4.2 Antennas. Antennas used to perform EMRH testing must convert the output of the transmitting equipment to an electromagnetic field which is repeatable and reasonably uniform over the test volume. As a rule, at frequencies below 1 GHz, the field intensity over the test item volume should not vary by more than 6 dB. At frequencies over 1 GHz, this is commonly not practical and the item must be moved in the field to ensure that all cracks, seams, and other penetrations are fully illuminated with the specified EMR environment.

F3.4.3 Field measurements. The field intensity must be measured using appropriate field measurement techniques. Field measurements should be made using equipment with an absolute accuracy of at least 2 dB. Either of the following techniques may be used to ensure a calibrated field measurement.

F3.4.3.1 Direct field measurement prior to test. One method of measuring the field intensity applied to a test item is to measure the applied field at the test location with the test item absent. With a field meter at the test location, raise the transmitter output until the desired test field level is reached. Measure the actual power output of the transmitter using a directional coupler and power meter or some other technique capable of measuring high power level. Annotate the transmitter power which produces this field intensity at the test volume. For test objects of significant size, the field should be measured at various points within the test volume to assess the field uniformity. Continue this process for each frequency

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and polarization to be tested. During the test, set the transmitter output power to that level annotated during the calibration procedure.

F3.4.3.2 Relative field measurement prior to test. The field intensity applied to a test item may be determined based upon field intensity measurements made in the empty test volume prior to the test. With a field meter at the test location, raise the transmitter output until a predetermined field level is reached. Measure the actual power output of the transmitter using a directional coupler and power meter or some other technique capable of measuring high power level. Annotate the transmitter power which produces this field intensity at the test volume and calculate and record the total transmission system gain. For test objects of significant size, the field should be measured at various points within the test volume to assess the field uniformity. Continue this process for each frequency and polarization to be tested. During the test, the test field intensity can be calculated using the transmitter output power and the system gain. If using this method, care must be taken to ensure that all power measurement equipment is linear from the calibration level through the test power level.

F3.4.3.3 Direct field measurement during test. The most direct approach to determining the field intensity on a fuze item is by placing field probes in the test volume during the test. In this way, field intensity can be measured directly during the test. If this method is used, care must be taken to ensure that the field measurement equipment does not interfere with the field significantly, the field probes are not interfered with by the test item (for example, there is an apparent field intensification near the ends of cylindrical objects), and that the field probes are close enough to the test item to closely approximate the field on the test item. As the requirements are somewhat conflicting, this method requires technical judgments to be made.

F3.4.4 Fuze instrumentation. The fuze being tested must be instrumented to measure the response of the EEDs to the EMR environment. For EEDs utilizing bridgewires to initiate the explosives, the key parameter to be measured is current. Normally, the EED is replaced with a current sensor with similar characteristics to the EED bridgewire. Care must be taken when instrumenting a fuze so that the instrumentation provides an accurate measure of EED response without significantly affecting the result. The primary concerns are that the shielding integrity of the fuze not be altered by the instrumentation and that the instrumentation not form an additional inadvertent antenna with different characteristics than that of the fuze electronic circuits. Some acceptable methods are explained in the following paragraphs.

F3.4.4.1 Vacuum thermocouple instrumentation. This type of instrumentation utilizes a thermocouple mounted in close proximity to a bridgewire. The bridgewire and thermocouple are suspended in a vacuum to maintain temperature stability. The thermocouple, which is close enough to the bridgewire to respond to changes in its temperature, outputs a DC voltage proportional to the amount of energy dissipated in the bridgewire. These devices can be calibrated using DC current to determine the response to the rms current induced by RF. The technical challenge to be overcome with this type of instrumentation is to mount the sensor in place of the EED and get its DC output out of the fuze in a way that will not disturb the EMR characteristics of the fuze. The preferred method is

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through the use of fiber optics. The output of the thermocouple can be amplified and used to drive a Voltage Controlled Oscillator (VCO). The VCO output can then drive an LED which feeds the fiber optic output. In some instances, this instrumentation may not fit inside the fuze or munition. In those instances, the DC output of the thermocouples may be brought out of the fuze or munition to an instrumentation package using filtered feed through connectors and shielded twisted pair. Tests should be accomplished to ensure that the instrumentation external to the munition is not being interfered with by the EMR environment nor providing a significant additional antenna for the fuze.

F3.4.4.2 Fiber optic sensor instrumentation. There are some commercially available instruments designed specifically to overcome some of the pitfalls of EMRH testing of munitions. These make use of small current sensors with a direct fiber optic output. One technology makes use of a small dot of phosphor on the bridgewire. The decay time of the phosphor is proportional to its temperature. A fiber optic line is mounted in close proximity in order to illuminate the phosphor and subsequently to measure its decay time. The sensor can be calibrated for temperature rise vs. temperature. Another technology uses a small semiconductor device which responds to current. Bridgewire is run from the posts of an EED to this semiconductor circuit. A fiber optic is mounted to monitor the status of the semiconductor. A choice of fiber optic sensor technology will depend on the parameters being measured and the instrumentation sensitivity requirement for the test.

F3.5 PROCEDURE. The specifics of the test procedure will vary from test to test due to the uniqueness of the fuzes being tested. However, the basic procedure should closely resemble the following:

a. Place the instrumented fuze in the test area in the configuration to be tested.

b. Orient the fuze to the source antenna as prescribed in the test plan.

c. Turn on the transmitter tuned to the frequency specified for test.

d. Gradually increase the field intensity until either the specified EMR environment is reached or the instrumentation shows a response approaching its damage level.

e. When applicable, perform any necessary actions on the test items (attaching cables in preparation for launch, etc.).

f. Record the field level, frequency, polarization, test item orientation, configuration, and instrumentation response. If no response is detected by the instrumentation, record the minimum detectable level.

g. Repeat steps a through f for all configurations, orientations, polarizations, and frequencies in the test plan.

F3.6 ALTERNATE AND OPTIONAL TESTS. None.

F3.7 RELATED INFORMATION.

F3.7.1 Data analysis.

F3.7.1.1 Extrapolation. In many test facilities the fuze's EMR environment requirements cannot be generated over all frequency ranges. A commonly accepted practice is to measure the response of the EEDs at the maximum field capability of the test facility and

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extrapolate those measurements to the specified field intensity. In order for this extrapolation to be valid the following requirements apply:

a. The instrumentation must be sensitive enough so that its minimum detectable level when increased by the ratio of the required field to the test field is still less than the no-fire characteristics of the EED reduced by the appropriate safety factor from F3.3.2.

b. There must be some reasonable evidence that the response of the system is linear in the region between that at which the measurements are made and that to which they are to be extrapolated.

If these two conditions are met, EMRH test data can be gathered at levels readily available at most test facilities and extrapolated to high EMR specification levels.

F3.7.1.2 Data presentation format. There are two main methods of reducing, extrapolating, and presenting the results of EMRH tests. One involves the calculation of the highest field level at which the required safety factor (see F3.3.2) is maintained and comparing it to the EMR environment specification (see F3.2.2) for the item tested. The other method is used to calculate the safety factor obtained at the specified EMR environment and compare it to the required safety factor from F3.3.2.

F3.7.1.2.1 No-fire field intensity (NFFI) format. When data is calculated and presented in this format, the environment in which the item is safe and reliable is readily apparent. The No-fire field intensity is a calculated field intensity at which the current induced in the EED is equal to the no-fire characteristic of that EED reduced by the appropriate safety factor. The NFFI is calculated using the following equation:

$$NFFI = TFI \times SF \times NFL / ML$$

where:

NFFI = No-fire Field Intensity

TFI = Test Field Intensity

SF = Safety Factor (from F3.3.2) expressed as a decimal ratio

NFL = EED 0.1%, 95% confidence no-fire level (current or voltage)

ML = measured level (current or voltage). If no response is measured during testing use the minimum detectable level as ML

This method would be useful for a quick evaluation of a system if the required EMR environments were to change. It also lends itself extremely well to a graphical presentation of the data which shows the systems response with respect to frequency and with respect to the EMR environment specification in a manner clearly understandable by persons not familiar with this type of testing.

F3.7.1.2.2 Safety factor format. When data is calculated and presented in this format, the level of safety afforded at the EMR environment specification is readily apparent. The safety factor afforded at the EMR environment specified is calculated using the following equation:

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$$SF = (RFI \times ML) / (TFI \times NFL)$$

where:

SF = safety factor at the EMR environment specified

RFI = required field intensity

ML = measured level (current or voltage). If no response is measured during testing use the minimum detectable level as ML

TFI = test field intensity

NFL = EED 0.1%, 95% confidence no-fire level (current or voltage)

This method would be useful for a quick evaluation of a system if the safety factor requirements were to change for an EED for some reason. This method does not, however, lend itself to a graphical presentation of the data in a meaningful way.

F3.7.2 Electroexplosive devices (EEDs). In order to perform an EMRH test on a fuze, a basic understanding of different types of EEDs is required so that instrumentation can be used which will measure the appropriate parameters and also so that data can be analyzed correctly. The following basic types of EEDs are often found in fuzes and their peculiar characteristics should be understood by the test designer.

F3.7.2.1 Hot bridgewire devices. Perhaps the most common type of EEDs in fuzes is the hot bridgewire. Most piston actuators and microdetonators are of this type. This type of EED has a small explosive charge mounted on a thin resistive bridgewire. When sufficient current passes through the bridgewire (either intentionally or unintentionally), the joule heating effect causes the charge to ignite. This, in turn, ignites the rest of the energetic materiel in the EED. These devices are normally initiated by a small voltage on a relatively large capacitor. For pulse modulated fields hot bridgewire devices respond to the average current level as opposed to the peak level unless sufficient energy is contained in a single pulse to initiate the device. The thermal time constant of the device should be considered with respect to the duty cycle and pulse repetition rate when evaluating a hot bridgewire device against a specific threat environment.

F3.7.2.2 Conductive mix devices (carbon bridge). These EEDs are initiated by an explosive mix which contains conductive materiel. As voltage is applied to the leads of this type of device, very little current flows until a threshold is reached where the current can jump between conductive particles. At this point the surge of current ignites the explosive mix which then ignites the remainder of the energetics in the EED. These devices are normally fired by a relatively high voltage from a small capacitor. These devices definitely are susceptible to peak voltages created by pulse modulated fields. They also pose unique instrumentation pitfalls as they are very non-linear and their impedance changes drastically as their firing voltage is approached.

F3.7.2.3 Exploding foil initiators (EFI). These detonators are used mainly in fuzes with an uninterrupted explosive train. This type of EED has a small conductive metal foil bridge which is housed underneath a plastic layer. When a sufficient current pulse is passed through the bridge it vaporizes, throwing the plastic "flyer" into a relatively insensitive

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explosive charge with enough energy to cause it to detonate. Their chief characteristic is that both a high voltage and high current are required with a fast rise time and short duration in order to fire an EFI. They are normally considered insensitive to EMR and are therefore the usual choice for Electronic Safe and Arm Devices (ESADs). It is, however, within the realm of feasibility for the high peak current and sharp risetime induced by some radar environments to initiate an EFI. For this reason an EFI cannot be ignored during an EMRH test and evaluation effort. Particular attention must be paid to determining the response of the device at the high peak field produced by radars and comparing them to the EFI's firing characteristics.

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